



NEUROLENS A MULTIMODAL AI FRAMEWORK FOR EARLY PARKINSON'S DISEASE DETECTION

Group ID: 25-26J-109

PROJECT SUPERVISION

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gettyimages

The silent crisis of Parkinson's Disease affects over 8.5 million people worldwide, and its prevalence is rapidly increasing.

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Problem Statement & Stakeholders

Parkinson's Disease (PD) is a progressive neurodegenerative disorder affecting over 10 million people worldwide.

The Problem

Over **70%** of Parkinson's Disease cases are diagnosed only after noticeable symptoms appear, missing the critical early detection window.

Impact

Late diagnosis delays treatment, increases healthcare costs, and reduces the effectiveness of early disease management.



Stakeholders



- Patients
- Healthcare providers
- Caregivers

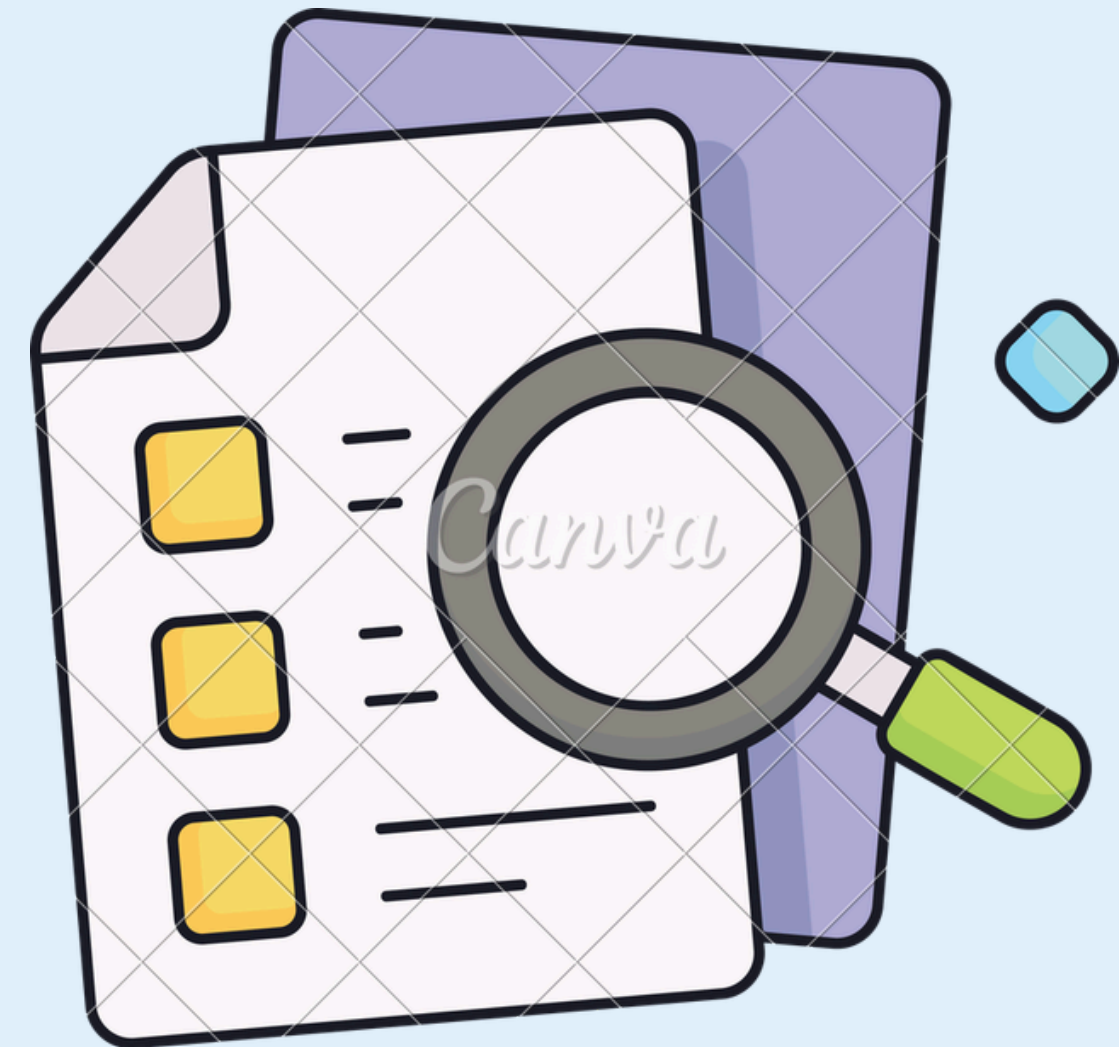
Research Gap & Objectives

Existing Research:

- Focus on single modality (only speech, only gait, or only handwriting).
- Some studies, limited to raw data → cloud (privacy & bandwidth issues).

Main Objective

- Develop a multimodal AI framework for early detection of Parkinson's Disease using multiple digital biomarkers.



Solution Overview

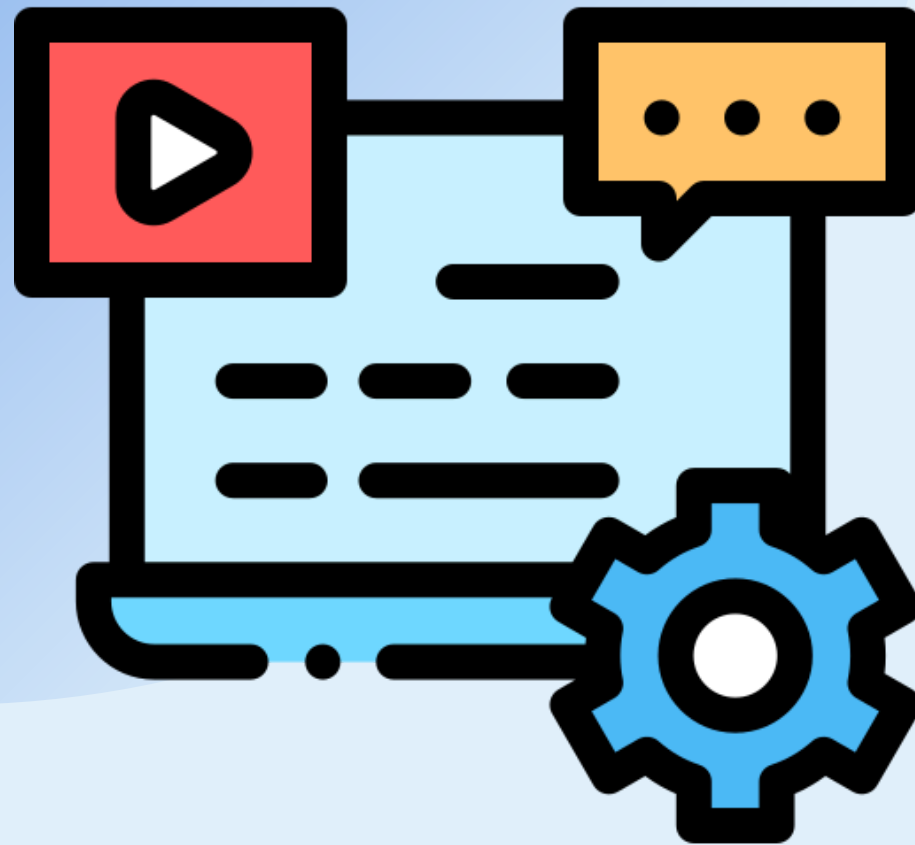
multimodal AI-powered mobile platform designed to enable accessible and early screening of Parkinson's disease through digital neurological assessments and intelligent risk prediction.

Key Features :

- Multimodal analysis (voice, motor, cognitive, wearable data)
- Real-time AI risk prediction
- Trilingual mobile interface
- Longitudinal health tracking



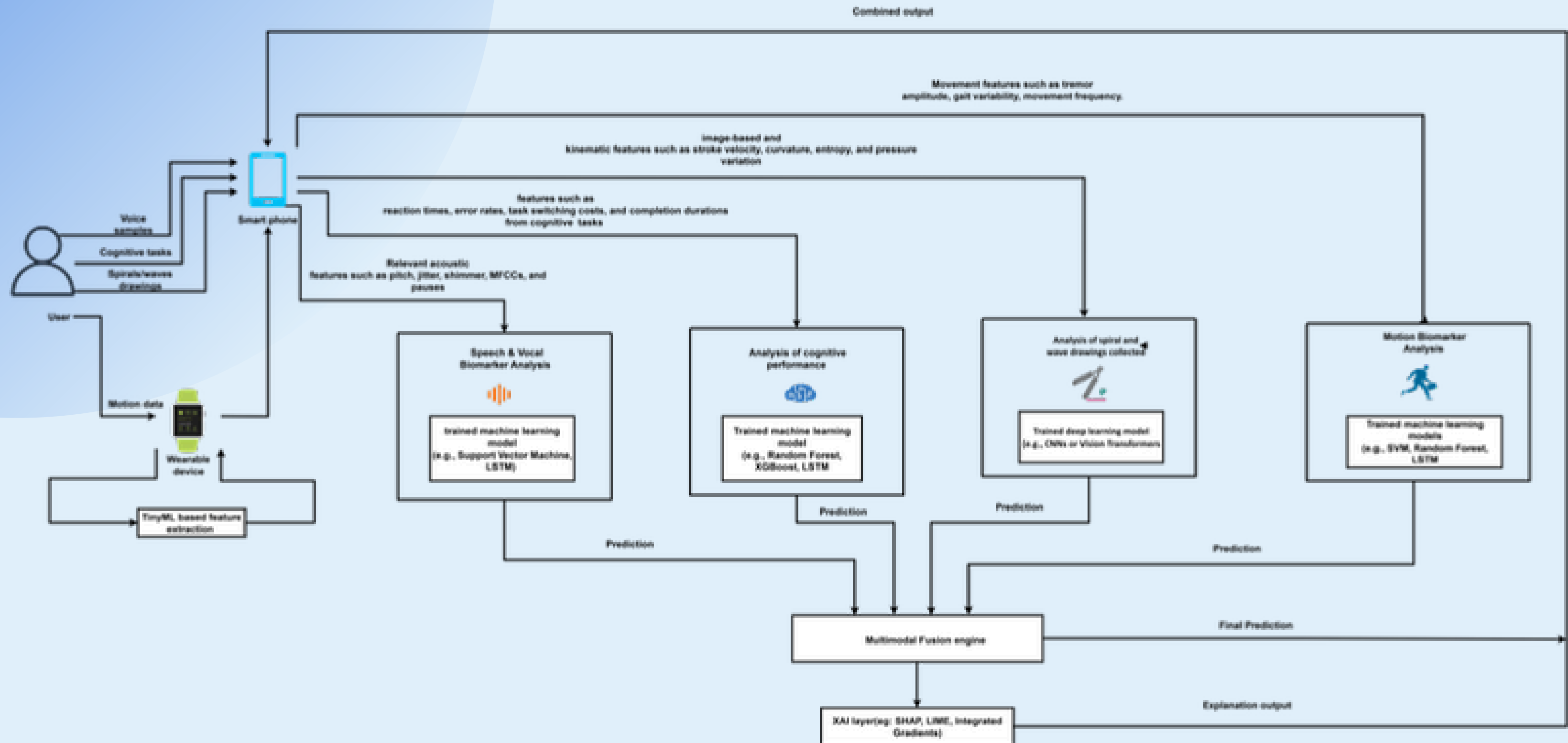
Solution Overview



Non-functional Focus:

- Privacy
- Scalability
- Interpretability
- Real-time analysis.

System Architecture & Integration



System Architecture & Integration

Component 1 – Wearable Motion Analysis

IT22093332 - Pathirathana W.N.G.R

- Analyze accelerometer and gyroscope data
- Detect tremors and gait abnormalities

Component 2 – Cognitive Biomarker Analysis

IT22551498 - Savithma P.T.C

- Analyze digital cognitive tasks
- Extract reaction time and error patterns



System Architecture & Integration

Component 3 – Speech & Vocal Biomarkers

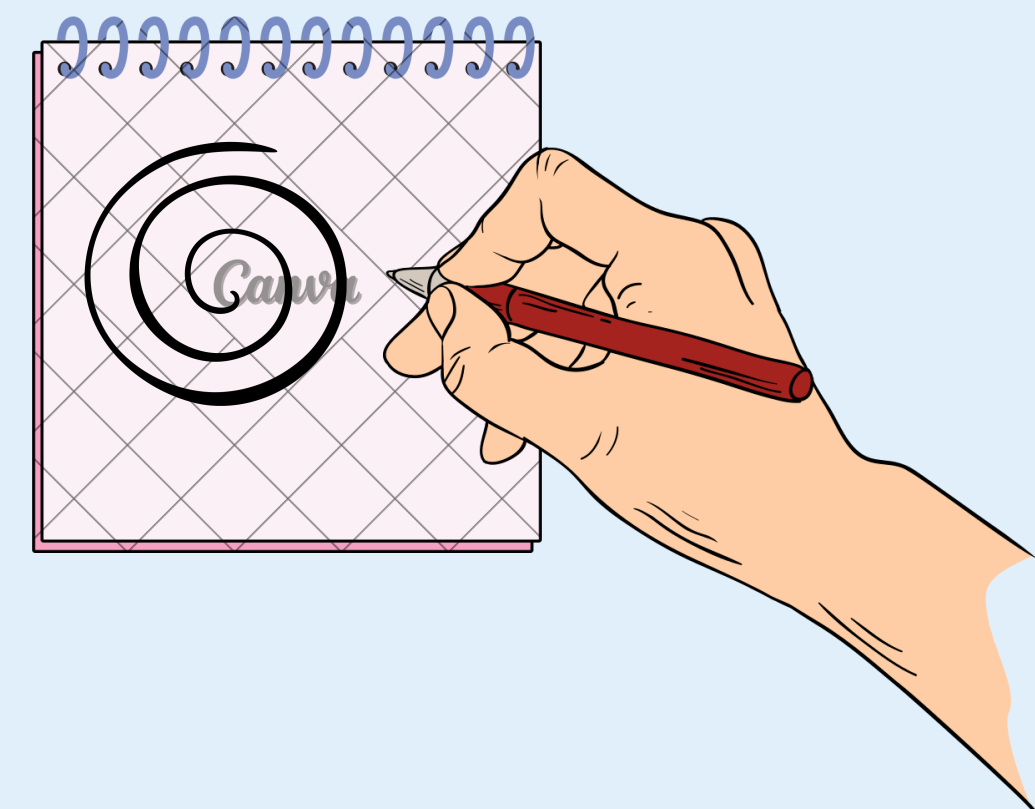
IT22079336 - Atigalage Y.V

- Analyze voice recordings
- Extract acoustic and linguistic features

Component 4 – Drawing Analysis

IT22884060 - Sriharan S

- Analyze spiral and wave drawings
- Extract motor control features



System Architecture & Integration

Integration points :

- **FastAPI** Framework: High-performance, modular REST endpoints for all modules:

POST /voice-analysis/predict_voice

POST /drawing-prediction/analyze

POST /cognitive-analysis/predict

POST /wearable-prediction/analyze



Deployment & Runtime Setup

- Containerized Architecture: Fully packaged with Docker for consistent local and cloud execution.
- Serverless Infrastructure: Optimized via **Google Cloud** for automatic scaling, high availability, and low-latency performance.

COMMERCIALIZATION

Free Tier

LKR 0

Forever Free

- ✓ Basic assessments
- ✓ Limited Features
- ✓ User acquisition focus
- ✓ Upsell opportunity

START FREE

Premium Tier

LKR 1,990

per month

- ✓ Full Feature Access
- ✓ Wearable Integration
- ✓ Real Time Monitoring
- ✓ Explainable AI (XAI)

UPGRADE NOW

Customer Persona

- Primary: Adults 40+ at risk for neurological conditions.
- Secondary: Caregivers, clinicians

Market Size

- Initial focus on Sri Lanka
- Global expansion potential as a scalable digital health solution

What is Unique ?

- Multimodal analysis (voice, motor, cognitive, wearable data)
- Real-time monitoring with digital biomarkers
- Trilingual mobile interface

COST & FEASIBILITY

Initiative	Amount (LKR)
Labor: Software development, AI training, UI/UX design.	6000
Material: Hardware	7000
Material: Cloud infrastructure, API licensing, servers	20000/yearly

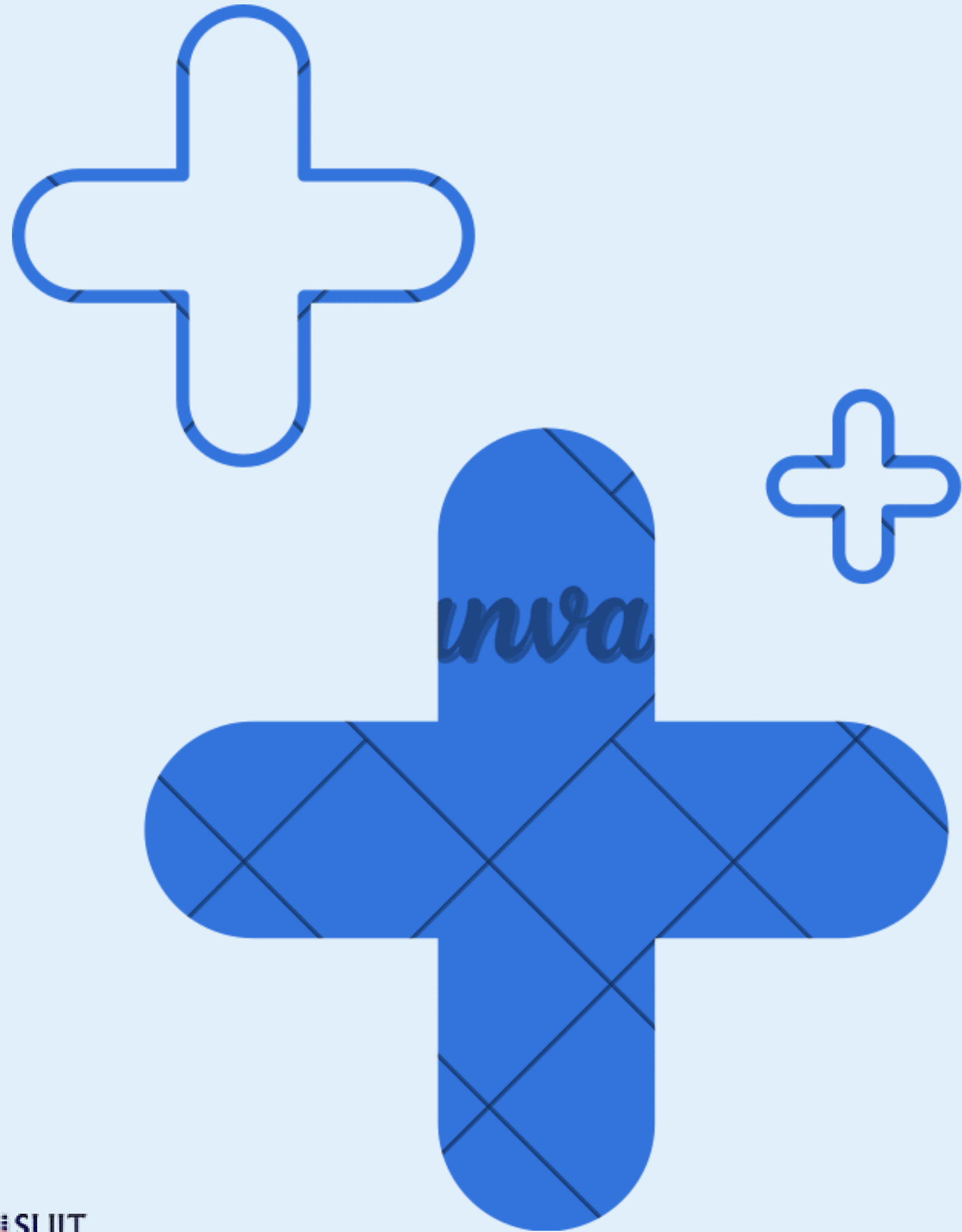


NeuroLens



DEMO





**ANY
QUESTIONS ?**





THANK YOU

